From-Scratch CNN Classification, Pure-Noise GAN Augmentation, and CycleGAN Style Transfer

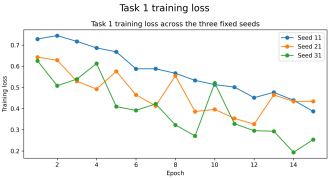
Qifeng LIU

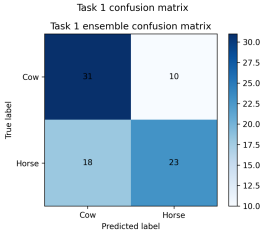
Dataset and Project Objective

- Three tasks:
  - Task 1: classify cows vs horses with a CNN trained from scratch
  - Task 2: improve classification with GAN-based data augmentation
  - Task 3: perform unpaired style transfer between the two classes
- Dataset size:
  - 82 training images and 82 test images
  - 41 cows and 41 horses in each split
- Main difficulty:
  - one training instance per class
  - unseen test instances
  - shared viewpoint grid
- Main objective:
  - improve the weak horse class without making the classifier unstable

Task 1: CNN Design

- Compact from-scratch CNN
  - 3 blocks of `Conv + BN + ReLU + MaxPool`
  - global average pooling
  - small fully connected head: $64 \rightarrow 32 \rightarrow 2$
- Why small?
  - dataset is tiny
  - overfitting is the main risk
- Training choices
  - class-weighted cross-entropy loss
  - cow:horse weight ratio $= 1 : 5$
  - 3 fixed seeds
- Final decision rule
  - average logits across the 3 seed models
  - then take the final class prediction

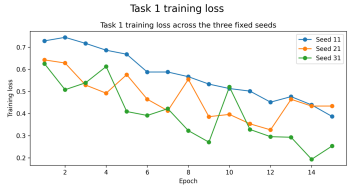


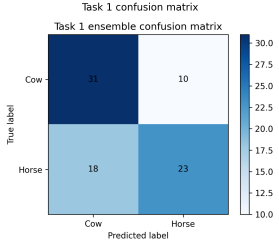


Task 1: Results and Key Point

Final Task 1 Ensemble

Metric	Value
Overall accuracy	65.85%
Cow accuracy	75.61%
Horse accuracy	56.10%



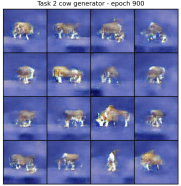


- The CNN already works well from scratch.
- The most important stabilizers are:
  - class-weighted cross-entropy loss
  - cow:horse weight ratio $= 1 : 5$
  - logit-level ensemble
- The remaining weakness is horse recognition.

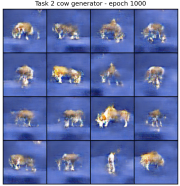
Task 2: Pure-Noise GAN Architecture

- Generator
  - input latent vector $z \in \mathbb{R}^{100}$
  - projection to feature map, then progressive `Upsample + Conv2d`
  - `BatchNorm + LeakyReLU`, final `Tanh`
- Discriminator
  - stride-2 convolution blocks
  - `LeakyReLU` and dropout
  - binary real/fake prediction
- Class-specific training
  - one GAN for cows
  - one GAN for horses
  - valid generated images are exported for manual selection

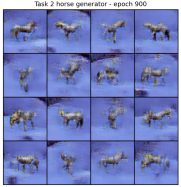
Cow generator at epoch 900



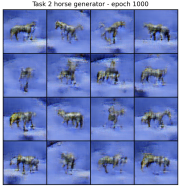
Cow generator at epoch 1000



Horse generator at epoch 900

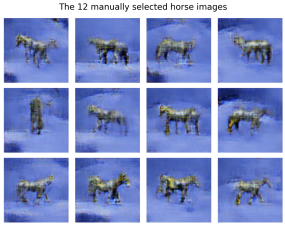


Horse generator at epoch 1000



Task 2: Final Augmentation Result

- Final manual selection
  - selected cows = 0
  - selected horses = 12
- Final result
  - 65.85% → 67.07%
  - cow unchanged
  - horse: 56.10% → 58.54%
- Key point
  - targeted asymmetric augmentation of the weak horse class
- Pretrained discriminator
  - rejected because horse generations failed visually

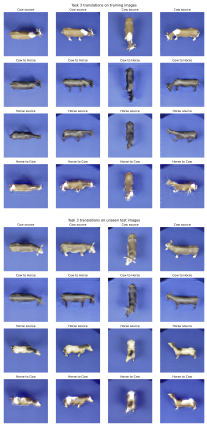


Task 3: CycleGAN Design

- Residual generators
  - encoder
  - residual bottleneck
  - decoder
- PatchGAN discriminators focus on local realism
- Training losses
  - adversarial loss
  - cycle-consistency loss
  - identity loss
- Replay buffer reduces discriminator oscillation
- Goal
  - change visual domain while preserving viewpoint and content structure

Task 3: Qualitative Result and Key Point

- The translations are evaluated qualitatively.
- Main visual criteria:
  - silhouette
  - viewpoint
- Result
  - plausible bidirectional translation
  - meaningful local style changes
- Key point
  - compact CycleGAN-style model is sufficient for visual transfer



Conclusion

- A compact from-scratch CNN was sufficient for this dataset.
- The key Task 1 choices were:
  - class-weighted cross-entropy loss
  - cow:horse weight ratio $= 1 : 5$
  - 3-seed logit-level ensemble
- Task 2 improved performance only after:
  - exporting valid GAN candidates
  - manually curating the final subset
  - augmenting the weaker horse class only
- Final result:
  - Task 1 $= 65.85\%$
  - Task 2 $= 67.07\%$
- The best system was not the most complex one.